Java 默写题（摘选自往年题）

Week1 & 2

最最基础内容

1. explain the purpose of each of the following Java tools

javac	Java compiler
java	the launcher program for Java applications
javadoc	the API documentation generator
jar	used to mange Java Archive (JAR) files.

2. list all the Java primitive data types

byte, short, int, long

float, double, boolean, char

3. use examples to explain the difference between x++ and ++x, in terms of their code behaviour?

post-increment example x++:

a post operation causes the variable to be used as is in the current statement and it is then incremented or decremented afterwards.

```
int a = 5;
int result = a+ a++;
```

after executing the statements above, result=10

pre-increment example ++x:

```
int a = 5;
int result = a+ ++a;
```

after executing the statement above, result = 11

4. why would someone use a break statement inside a switch?

5. 什么是 pass-by-reference 和 pass-by-value?

pass-by-reference: the address of the variable is passed to a method.

pass-by-value: the actual value of the variable is passed to the method.

6. 描述不同种类 comments 的 syntax 和 purpose

7. Which of the following is not valid Java identifier?

RESULT

12345

x12345y

black&white

answer_7

答案：

12345 an identifier can't begin with a digit

black&whit an identifier can't contain the character &

8. What is wrong with the code below? How to fix it?

code:

```
int length;
final int MIX_LENGTH;
if (length=MAX_LENGTH)      System.out.print("The length is minimal");
```

答案：

The if condition requires a boolean value

however length=MIN_LENGTH is an assignment which will result an int value

修改：length==MIN_LENGTH

9. explain the difference between while and do-while loops，然后给出 Java syntax

答案版本 1:

unlike the while loop, in a do-while, the condition is checked at the end of the first loop cycle;

so the code inside a do-while loop will run at least once.

答案版本 2:

After the first loop iteration, condition must evaluate to true in order for the statement inside the loop to be executed again

The statement in the do-while loop will be executed at least once, regardless of the initial value of the condition, whereas the statement of a while loop will never be executed, unless the condition evaluates to true.

```
while (condition){
    // statement
}

do{
    // statement
} while (condition);
```

面向对象

1. describe the difference between a Java object and a Java class.

2. how does an object relate to a class?

An object is an instance of a class.

Encapsulation

1. 什么是 encapsulation?

Encapsulation hides your data, protecting it and allowing it to only be changed by set methods, meaning that you have control over how it can changed.

2. How is encapsulation implemented in a Java class?

Encapsulation is achieved by marking variables as private.

3. 给出一个 implement encapsulation in Java at the point of declaring an instance variable 的例子

4. Is there any case where we would not encapsulate data?

If you have a constant value(indicated by the keyword final),
then it is not necessary to encapsulate it as that value will not change.

Javadoc

1. 生成 Example.java 文件的 javadoc comments 的指令是什么？

javadoc -d docs file.java 把这一句中的 file 的名字替换成 Example 即可：
javadoc -d doc Example.java

2. give one example of a javadoc tag, along with its description.

3. 什么是 javadoc？

4. javadoc 的目的是什么？

Access Modifier

1. Explain the meaning of access modification in Java, by comparing the private and public access modifiers, including any advantages or disadvantages.

2. What is the difference between the private, public, protected access modifiers?

private: it means that sth. is visible to (or within) the class only.

public: it means that sth. is visible to the world.

protected: it means that sth. is visible to (or within) the package and all subclasses.

3. How to access an instance variable if it is private?

4. We may store a fixed variable marked as public final static.
For example, a constant such as the absolute zero value of temperature would be written as "public final static double ABSOLUTE_ZERO = -273.14;"
Will making this public cause any problem?

5. Find the error in the code below and explain why the code doesn't compile.

```
public class TestClass{
    void doSomePrinting(){
        System.out.println("I Love Java");
    }
    public static void main(String[] args){
        doSomePrinting();
    }
}
```

答案：

calling the doSomePrinting() methohd is invalid

We can't call a non-static method from a static method

两种解决方法：

（1）面向对象，给类创建一个对象，然后再调用方法

Instantiate the class TestClass

access the instance method doSomePrinting().

代码：

```
TestClass myClass = new TestClass();
myClass.doSomething();
```

（2）把方法也弄成静态的，然后就可以直接调用

Declare the method doSomething() as static, and access it in the main() method without instantiation

this is not object-oriented, so it is not good programming practice
代码：
public static void doSomething()

Array and ArrayList

1. Give example code to create an array which is filled with the values 1,2,3,4,5 in this order

2. assume that you have a program that creates an array called numbers with 10 integer values. Give a brief description of what will happen if you attempt to run: numbers[15]=7

3. array 和 arraylist 有什么不同（至少 2 个）（不全面）

arrays have a pre-defined size, set when the array is declared

arraylist have a dynamic size, thus can grow and shrink

声明例子：

```
int [ ] myArray = new int[3];
ArrayList<Integer> myList = new ArrayList<Integer>( );
```

Inheritance

1. describe the concept of inheritance and state 2 of its benefits.

2. extends 关键字有什么用

3. 什么是 polymorphism? explain how that is achieved in Java，给出例子

4. 什么是 method overriding?

Method overriding is the practice of re-defining the implementation of a method defined in the superclass.

The method signature must be exactly the same as that of the superclass.

Method overriding allows a subclass or child class to provide a specific implementation of a method that is already provided by one of its superclasses or parent classes.

5. 什么时候使用 super 关键字 in relation to method overriding?

Any calls to an overridden method from within the subclass will by default invoke the version implemented in the subclass.

The method name can be preceded with "super", in order to explicitly invoke the superclass version of the method.

6. explain why you wish to override the "public boolean equals(Object obj)" method defined in the class "java.lang.Object"

The Object class implementation compares the identities of the calling object and of the object passed in as the argument,

i.e. it would only return true if both references are referring to the exact same object.

However, in practice it is more useful to be able to return true when 2 objects have equal state,

this is why we normally override the method equals( )

7. explain why you wish to override the "public String toString( )" method defined in the class "java.lang.Object"

The object class implementation returns a String consisting of the name of the class of which the object is an instance, the "@" character and the hashcode

However, in practice is more useful to return a string containing the state of the object, i.e. the values of the instance variables,

this is why we normally override the method toString( )

Abstraction

1. What is the difference between a concrete class and an abstract class?

答案：

An abstract class cannot be instantiated, whereas a concrete class can be instantiated.

Abstract classes are usually extended by one or more concrete classes.

2. 什么是 abstract 关键字？解释 2 种不同的用法

用法 1:

If a **class** is marked as abstract, it means that it can't be instantiated with the keyword new and it must be extended to make a concrete class.

用法 2:

If **method** is marked as abstract, then it must be implemented in an inheriting class.

Week3

(1) Interface

1. 解释 Comparable interface，解释为什么使用它和它的返回值。

2. 判断以下语句是否正确

(1) An interface is a collection of abstract methods. 对

(2) The functionality of an interface is fully specified. 错，只写了个名字，不实现

(3) It is possible for an interface to extend another interface. 对

(4) An interface cannot be instantiated. 对，是抽象的模版

(5) A class can only implement a single interface. 错，可以实现多个接口

(6) An interface can contain a method implementation.

不知道，有了默认方法和静态方法之后其实 interface 里面也可以写方法的具体实现

3. Describe what interfaces are.

答案：

An interface is like a certificate that says “I provide these services”

can’t make an instance of an interface

its entire functionality is left unspecified

a class implements an interface by using the keyword “implements”

the class must provide an implementation for each of the methods listed in the interface（只提供一部分的话就变成抽象的了）

4. State two similarities and two differences between interfaces and abstract classes

答案：

Similarity

can’t be instantiated

can contain abstract methods

are supertypes to classes that extend/implement them

Difference

Abstract classes can have concrete methods 写不写都行

Interfaces can only have abstract methods 不可以写具体代码

Interfaces support a kind of multiple inheritance

Abstract classes don’t

Abstract classes are extended

Interfaces are implemented

5. Explain why multiple inheritance is not supported for abstract classes, and how interfaces can be used to provide a kind of multiple inheritance.

答案：

Abstract classes can have concrete methods

multiple inheritance of abstract classes could result in ambiguity about which superclass method to call

Interfaces provide a kind of “multiple inheritance” by allowing to extend several interfaces

we can have interface A extends B,C (B,C are interfaces)

no interface methods are concrete, this doesn’t introduce ambiguity about which code to execute.

(2) GUI

1. 判断下面的语句是否正确，错误的需要解释为什么错了

(1) Only one layout manager can be used per container. 对，一个容器里就一个管理

(2) Swing's list component(JList) only display text. 错，图像也可以显示

(3) A mouse object contains a method called mouseDragged().

错，不是 object 有这个方法，应该是 MouseListener 接口有

(4) The JFrame constructor takes as input a String representing the title you will see at the top of the window. 对，可以有字符串作为构造函数的输入，作为标题

(5) The actionPerform is the only method of the ActionListener interface.

错，应该是 actionPerformed

(6) GridLayout is the default layout for the Jpanel class.

错，Jpanel 默认的应该是 FlowLayout

(7) GridLayout(2,4) arranges the components in 2 rows and 4 columns.

对，先行后列

2. Explain how the rows and cols parameters in GridLayout(int rows, int cols) affect the components' layout.

3. What are the 3 main concepts when doing GUI programming in Java?
Give a brief description of each concept.
Briefly describe how the Three concepts are used together when designing a GUI.

4. Describe the purpose of layout managers in GUI programs

答案：

Layout managers provide a way to format components on the screen in a platform independent way

5. Describe the main difference between a GUI interface and a command line interface

答案：

A GUI interface is a graphical user interface that allows users to interact with a computer using widgets like buttons, icons and windows.

A command line interface allows users to interact with a computer but only via typing in commands.

6. Explain the differences between containers and components. Include an example of both a container and a component.

答案：

A container is a widget that has containment abilities

It is used to hold and group other widgets such as textboxes, radio buttons etc.

example: JFrame, Jpanel

A component is a widget that has no containment abilities
It is atomical of graphical widgets such as labels, text areas, buttons etc.
example: JButton, JLabel

7. Describe the concept of layout manager and indicate two examples of Java layout managers.

8. Describe the three steps required to implement an event handler for the button.

(1) Implement the ActionListener interface

In the declaration for the event handler class, code specifies that class either implements a listener interface or extends a class that implements a listener interface.

(2) Register with the widget

Code indicates that your program wants to listen for events, by registering an instance of the event handler class as a listener upon one or more components.

(3) Define the event-handling method

Code implements the methods in the listener interface.

(3) Memory management

1. when is an object Not marked for garbage collection. why is this the case?

2. when is an object marked for garbage collection. why is this the case?

3. explain the differences between Local and Instance variables in terms of where they are declared and how they are initialized

4. What are the two main areas of memory in Java?
Give an example of what might be stored in each area.

5. When is an object marked for garbage collection?
答案:
When there is no reference to it.

6. What is a null reference?
答案：
a reference that doesn't refer to any object

7. What is wrong with the code fragment below?
How to fix it?
By the end of the code, the Dog called "Bob" isn't eligible for GC, explain why.
How to make it eligible for GC?
code:

```
Dog myDog;
myDog.sleep();
```

答案：
（1）myDog can't invoke the sleep method
because it has no reference to any Dog object
（2）改成 Dog myDog = new Dog();
（3）the Dog object called "Bob" still has a reference to it.
（4）myDog = null;

8. Describe three different ways to make an object eligible for Garbage Collection. Include one illustrative example for each of the described approaches.

答案：

（1）set object to null

```
Object myObj = new Object();
myObj = nulll;  // myObj is now ready for GC
```

（2）the object reference is assigned to another object and there are no other reference variables that refer to it

```
Object myObj = new Object();
myObj = new Object();
//first created instance of Object is no longer referenced and is now ready for GC
```

（3）the object reference goes out of scope permanently

```
for(int i = 0; i<5; i++)
    Object myObj = new Object();
// myObject is ready for GC after each iteration, and at the end of the loop
```

9. Do you need to deallocate memory when an object is no longer in use? If yes, how? If not, why?

答案：

It is not necessary to do this, because Java has Garbage Collection which collects objects that are no longer in use, and destroys them. This means that unused memory is automatically released.

10. 什么是 scope of a local variable? give a short coding example demonstrating the scope of a local variable.

a local variable has block scope, which means within the brackets{ } only

例子：

```
if(a==2){
    int b = 5;
    }
```

variable b cannot be accessed outside of the if statement.

11. local variable 和 instance variable 之间有什么主要的区别？

local variables have a block scope, whereas **instance variables** have an object scope(depending on the access modifiers used).

local variables need to be initialised with a value before used, whereas **instance variables** will be assigned with a value automatically if a value is not provided.

12. local variable 和 instance variable 之间有什么主要的区别？

Instance variables are declared inside a class but not inside a method

they are initialized to the default value

they are valid (have scope) throughout the entire class

Local variables are declared within a method

they are not initialized to the default value (they must be initialized)

they are valid (have scope) only within that method

13. How do you refer to an instance variable when a local variable with the same name exists within the same scope? give a short example.

You can refer to the instance variable by using the this keyword.

例子：String name;

```
public void setName(String name){
    this.name==name;
}
```

(4) Strings

1. Explain the purpose of the equals() method in the String class.

答案：

It compares the content of the Strings, rather than the memory references.

(== compares the memory references, rather than the content of the Strings）

2. explain the meaning of each of the four lines and the context in which each line of code is used.

this();

this.toString();

super();

super.toString();

Week4

(1) Numbers

1. Explain the key difference of the static and final Java keywords in terms of initialisations, modifications and methods.

2. 比较 primitive data type 和 wrapper data type

	primitive int i	wrapper Integer i
Give the syntax, showing how you declare it with the value 0		
Explain what is happening with each stage of the code		
How is it initialized?		

3. 解释下面 4 行代码中的最后两行代码在运行的时候会做什么

```
int a = random.nextInt(13) + 1;
double b = random.nextDouble() + 4.0
System.out.println(a);
System.out.println(b);
```

4. Describe how recursive algorithms are implemented in Java, using the cod below as a reference example. Refer to specific lines of code in your answer.

Code:

```
public int recurse(int x){
    if(x==0){return x;}
    else{return x + recurse(x-1);}
}
```

5. 什么是 recursion

答案：

Recursion is a technique in which a method calls itself a number of times to solve a problem.

6. What is a class variable? When would you use it?

答案：

A type of variable that is created when its class is created, rather than when an instance of the class is created. There may be *n* instances of the class, but there will only be one instance of a class variable. Class variables are also referred to as static variables, and the modifier **`static`** is used to declare them.

Example of when to use a class variable: when we want to count how many instances of a class are made.

7. What abstract class do all numerical wrapper classes in Java extend?

答案：

the abstract class Number.

8. Name two wrapper classes. Write the necessary code to instantiate each of them.

答案：

Byte, Integer, Double, Short, Float 都行

Integer i = new Integer(10);

9. Provide an example of a case when a wrapper class is necessary.

答案：

When adding a primitive data type in an ArrayList.

因为 ArrayList 里面只能放 object，所以往里面存数字的时候需要转换成 Integer

但是现在有了自动装箱，其实我们平时也不太会注意到这个问题

(2) Exception

1. Create a user-defined exception class called InvalidDataTypeException.
Write the minimum amount of code that would be necessary to do this.
Indicate the name of the Java file that you would put that code in.

2. Write a user defined exception class called UserNotFoundException.
Write the minimum amount of code that would be necessary to do this.
Don't need to write constructor methods.

3. 判断下列语句是否正确
It is possible to have a try block without a corresponding catch block.
正确的，可以有一 try 但没有 catch，但必须有一个 finally
finally 块包含的代码无论是否发生异常都会被执行。

4. When handling multiple exceptions in Java, does the order of the catch blocks matter?

5. If you are given a Java program that uses exceptions to check if the denominator of the fraction is equal to zero.
What would be an alternative way to perform the same check?
What would be an advantage of this alternative way?

6. Whenever performing I/O in Java, it is necessary to use try and catch statements. Explain why this is.

7. Describe the concept of exception in Java.
答案：

an exception is an object and indicates that something out of the ordinary can happen when a program executes.

provide both a notification of errors and a way to handle them.

8. What are exceptions used for?
答案：

Exceptions provide both notification of errors and a way to handle them.

9. What is the output of the program below when assertions are disabled and when assertions are enabled?

Code:

```
public class AssertionTest{
    public static void main(String[] args){
        try{
            assert(false);
        }
        catch(Exception ex){
            System.out.println("Caught an exception!");
        }
        catch(Error err){
            System.out.println("Caught an error!");
        }
    }
}
```

10. Describe the purpose of try, catch, throw, and throws. Include an example illustrating where each of those 4 keywords are used.

答案：

try: declares the block in which normal code execution is attempted, if any statement can't be executed successfully, then the program's control junps to a catch statement if one exists.

catch: defines how to handle the exception when it occurs.

throw: a statement that causes an exception to be thrown.

throws: declare that a method "throws" an exception and it may result in an error if used in a particular way

例子：

```
public void exampleMethod() throws Exception{
    if( // an error occurs ){
        throw new Exception();
    }
    try{
        // some statements
    }
    catch( Exception e ){ }
}
```

(3) File I/O

1. There are 2 types of I/O streams: character and byte streams. Explain why they are both useful and name one of the studied character streams.

Character streams are useful for working with data in human-readable format, that will need to be dealt with directly by a person.

例子：FileReader, FileWriter

Byte streams are useful when the data is expressed in machine-readable format and will only be used internally by a computer.

translation of the data is not necessary, so it results in a more efficient process than using a character stream, which does require data translation.

2. Describe the concept of stream.

答案：

A stream is a connection to a source of data or to a destination for data.

Streams can represent any data, so a stream is a sequence of bytes that flow from a source to a destination.

3. Name the types of stream available in Java.

答案：

In a program, we read information from an input stream and write information to an output stream.

2. 判断以下语句是否正确

A Java program can't manage more than ten streams simultaneously.

错误，可以同时管理很多，不限制同时管理几个流

(4) Collection

(5) Sorting